1. Networking

- OSI & TCP/IP Models
- IP addressing, Subnetting, CIDR
- DNS, HTTP/HTTPS, FTP, SSH
- TCP vs UDP
- Routing and Switching basics
- Socket Programming (intro with Python or C++)

2. Operating Systems

- Process vs Thread
- Scheduling algorithms (Round Robin, FCFS, SJF)
- Memory Management (Paging, Segmentation, Virtual Memory)
- Deadlocks, Concurrency, Synchronization
- File Systems
- System Calls
- Linux fundamentals (bash, permissions, file system hierarchy)

3. Computer Architecture

- Number systems and binary arithmetic
- Instruction Set Architecture (ISA)
- CPU Organization (ALU, Registers, Control Unit)
- Memory hierarchy (Cache, RAM, HDD/SSD)

- Pipelining, Caching
- RISC vs CISC
- Microprocessors vs Microcontrollers

4. Clean Code & Software Architecture

Clean Code Principles

- Naming conventions
- Functions: small, single responsibility
- DRY, KISS, YAGNI
- Code smells and refactoring
- Comments and documentation

Design Patterns (intro level)

- Singleton, Factory, Observer, Strategy
- MVC (Model-View-Controller)

SOLID Principles

- Single Responsibility Principle
- Open/Closed Principle
- Liskov Substitution Principle
- Interface Segregation Principle
- Dependency Inversion Principle

Software Architecture

- Layered Architecture

- Hexagonal Architecture (Ports and Adapters)
- Microservices vs Monoliths
- RESTful APIs basics

5. Programming Practice

- Language: Python/Java/C++ (pick one as primary)
- Version Control (Git basics, GitHub)
- Writing unit tests (JUnit / pytest / unittest)
- Building small projects (CLI app, REST API)

6. Databases

- Relational Databases (MySQL/PostgreSQL): CRUD, Joins, Normalization
- NoSQL basics (MongoDB, Redis)
- Transactions, ACID properties
- Indexing and Query Optimization basics

7. Cloud & DevOps

- Basics of cloud platforms (AWS/Azure/GCP) – EC2, S3
- Containers (Docker fundamentals)
- CI/CD pipelines basics
- Deployment of a sample web app

8. Tools and Ecosystem

- IDEs (VS Code, IntelliJ)
- Debugging techniques and tools
- Package managers (npm, pip)
- Static analysis tools (ESLint, SonarQube)